Deep Learning on Text Data

Recruitment & HR | NLP & Machine Learning

Github Repo: https://github.com/nisansalasandu/recruitment-nlp-lab

Project Report

Name	**Ruwan Pathiranage Sanduni Nisansala**
Project Title	Neural Network-Based Resume & Job Description Analysis
Focus Area	Recruitment & HR / NLP & Machine Learning
Objective	Apply neural networks to analyze resumes and job descriptions
Frameworks	TensorFlow / Keras, scikit-learn, NLTK, Python

1. Executive Summary

This report documents a complete NLP and deep learning pipeline applied to recruitment data for Gamage Recruiters. The project processes raw resume and job description text data through a series of structured notebooks — from exploration and cleaning, to keyword extraction, neural network classification, and cosine similarity-based candidate–job matching.

The pipeline demonstrates that basic feedforward neural networks (MLPs) trained on TF-IDF features can reliably classify resumes by domain category and infer experience levels. A shared TF-IDF vectorizer further enables scalable candidate–job matching across 23.3 million pairs.

Component	Result
Resume Dataset	2,484 resumes | 4 categories | 466 after domain filtering
Job Description Dataset	1,615,940 rows | sampled to 50,000 for processing
Category Classifier	MLP (512→256→128→4) trained on 5,000 TF-IDF features
Experience Classifier	MLP (512→256→128→4) with class-weighted training
Matching Scale	50,000 jobs × 466 resumes = 23.3 million pairs
Mean Top-1 Match Score	0.1968 (unfiltered) | higher with category filtering
Strong Matches (score >0.30)	Reported in notebook 05 results

2. Data Overview

2.1 Resume Dataset

The resume dataset contains 2,484 resume records across 25 job categories. For this project, 4 target domains were selected for classification and matching tasks.

Data Source –

Resume Dataset: http://kaggle.com/datasets/snehaanbhawal/resume-dataset

Job Dataset: https://www.kaggle.com/datasets/ravindrasinghrana/job-description-dataset

Column	Type	Description
ID	Integer	Unique resume identifier
Resume_str	Text	Full resume text (avg 811 words)
Resume_html	Text	HTML-formatted version (not used)
Category	String	Job domain label (25 categories)

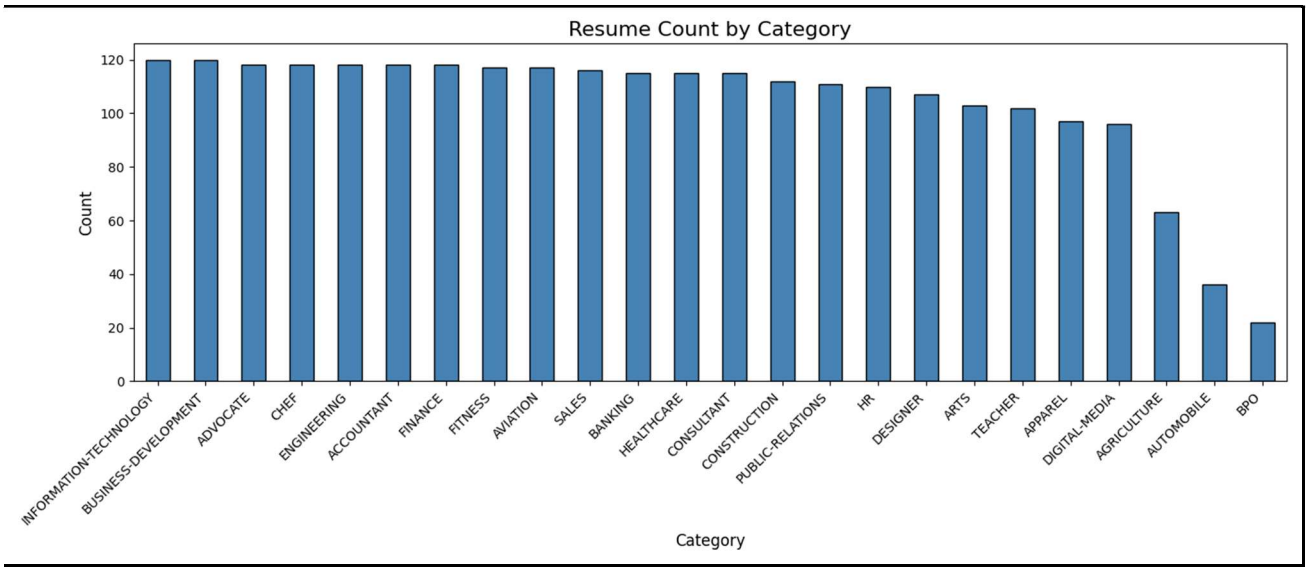

Figure: Resume Count by Category

2.2 Job Description Dataset

The job descriptions dataset is significantly larger with 1,615,940 rows and 23 columns. Processing was performed on a random 50,000-row sample for computational feasibility.

Key Columns Used	Description
Job Description	Primary description text (avg 24 words — very short)
skills	Required skills list
Responsibilities	Job responsibilities text
combined_text	Enriched field: description + skills + responsibilities
Job Title	Title of the job posting
Role	Role category

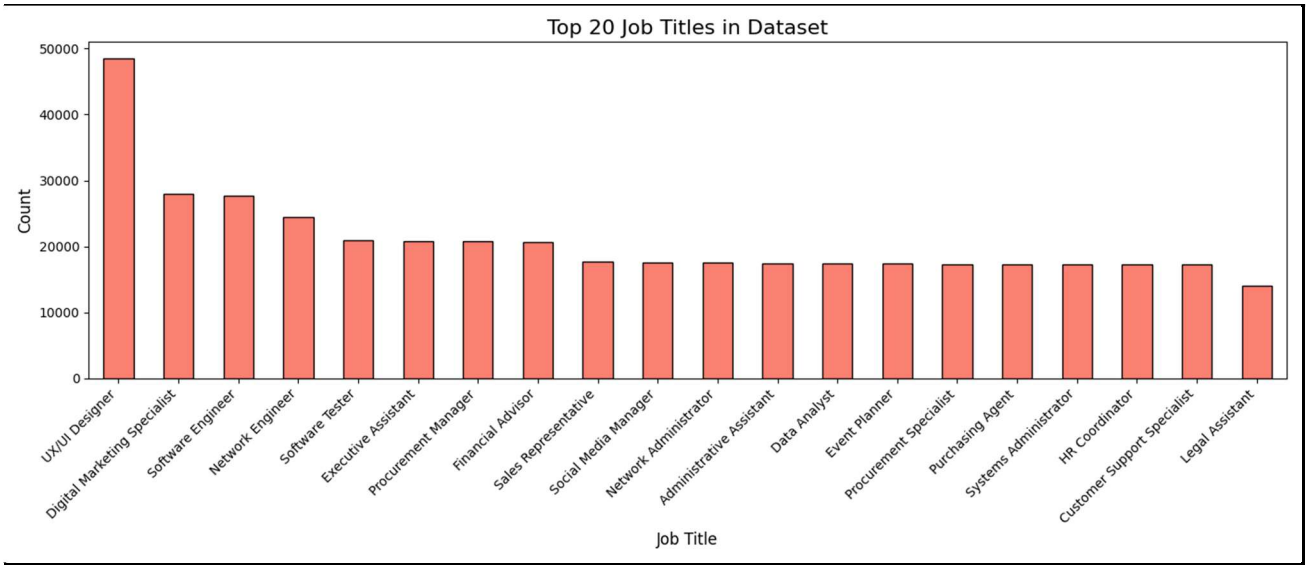

Figure: Top 20 Job Titles in Dataset

3. Text Preprocessing (Notebook 02)

All text data was cleaned through a 6-step pipeline before being fed into any model. This step is critical as raw resume text contains HTML artifacts, special characters, inconsistent casing, and noise words that would mislead neural networks.

Step	Action	Purpose
1	Handle null / missing values	Prevent downstream errors
2	Convert to lowercase	Uniform token comparison
3	Remove special characters, numbers, punctuation	Reduce noise
4	Tokenize with NLTK word_tokenize	Word-level units
5	Remove NLTK English stopwords	Eliminate non-informative words
6	Lemmatize with WordNetLemmatizer	Normalize word forms

3.1 Domain Filtering

Resumes were filtered to 4 target domains to keep the classification task focused and balanced:

Domain Label	Resume Category	Count After Filtering
HR / Management	HR	110 resumes
Finance / Accounting	ACCOUNTANT	118 resumes
Engineering	ENGINEERING	118 resumes
IT / Software	INFORMATION-TECHNOLOGY	120 resumes

3.2 Job Description Enrichment

Since the Job Description column averaged only 24 words — too short for meaningful NLP — a combined_text field was created by concatenating Job Description + skills + Responsibilities. This increased the average word count significantly and improved vectorization quality.

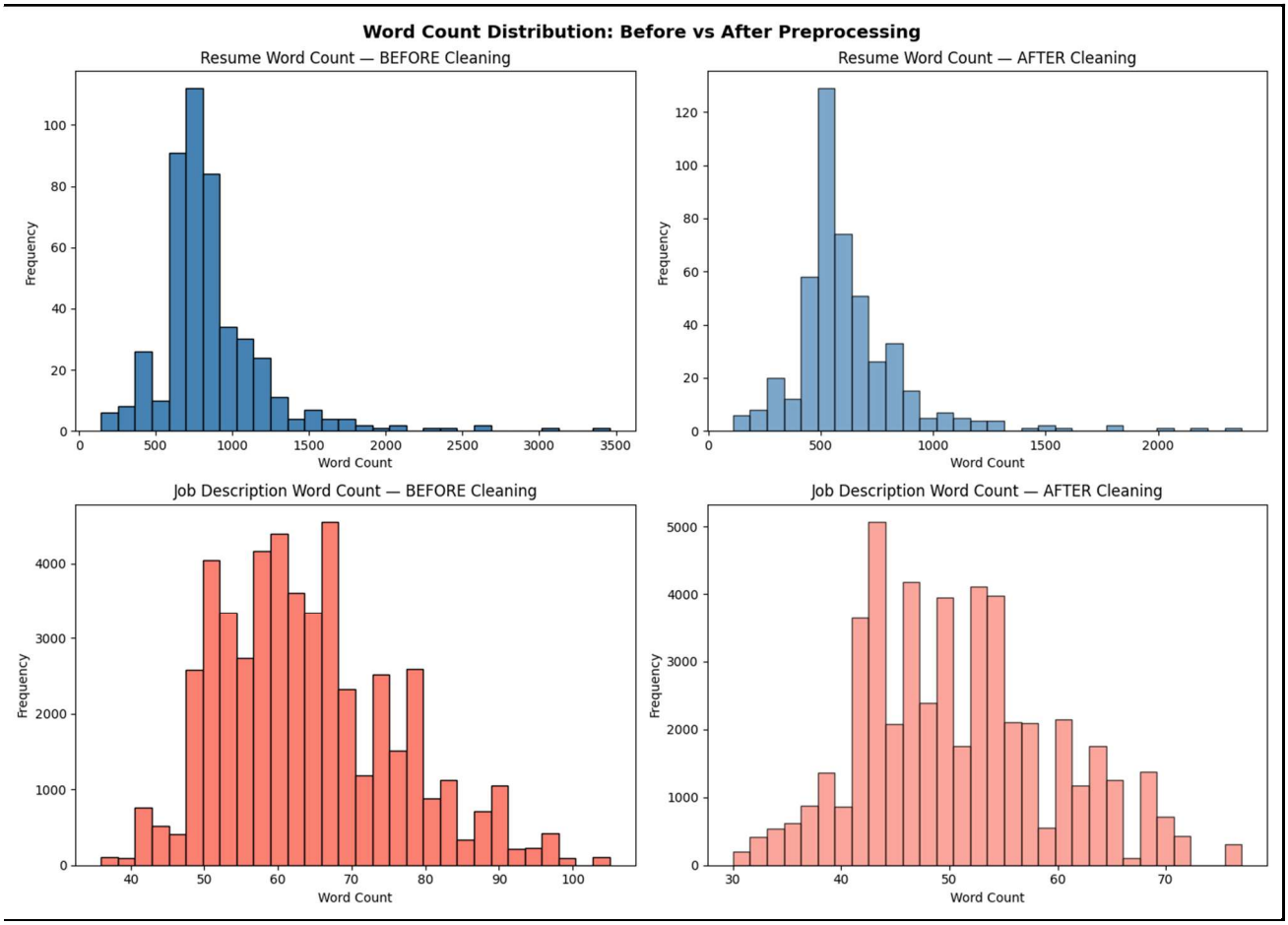

Figure: Word Count Before vs After Preprocessing

4. Keyword Extraction — TF-IDF (Notebook 03)

TF-IDF (Term Frequency–Inverse Document Frequency) was used to identify the most important and representative keywords in both resumes and job descriptions. Words with high TF-IDF scores appear frequently in a specific document but rarely across all documents, making them distinctive markers.

Parameter	Resume TF-IDF	Job TF-IDF
max_features	5,000	5,000
ngram_range	(1, 2)	(1, 2)
min_df	2	3
Matrix shape	466 × 5,000	50,000 × 5,000

4.1 Top Keywords Per Resume Category

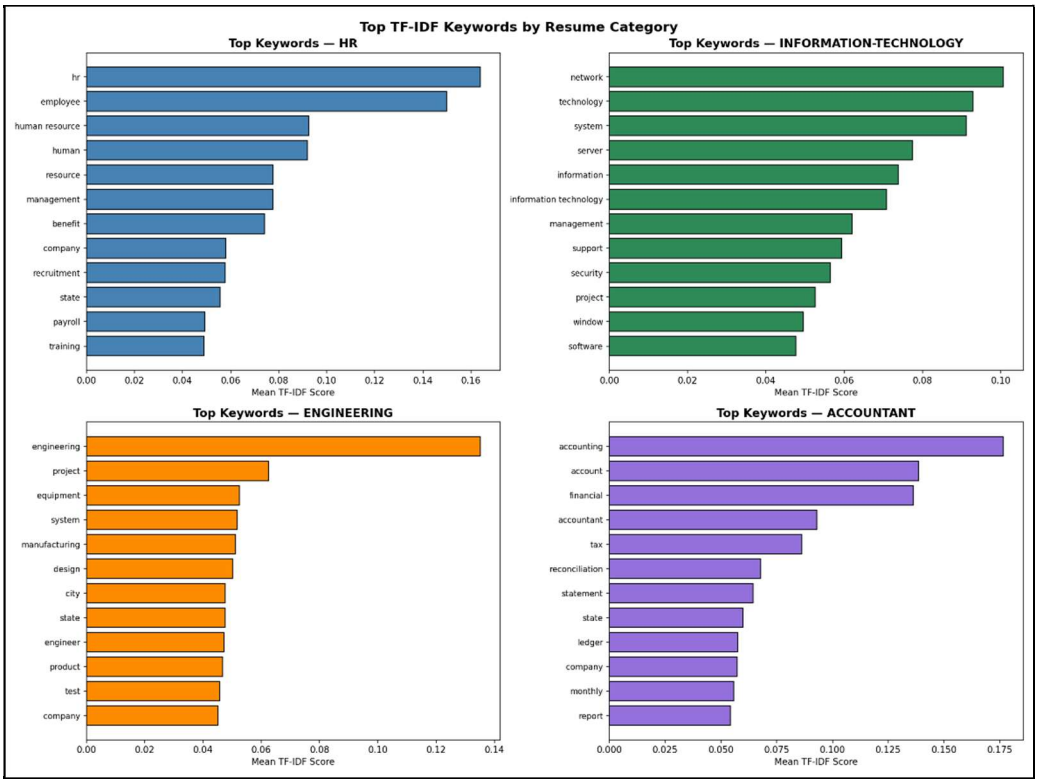

Figure: Top 12 TF-IDF Keywords Per Resume Category

4.2 Word Clouds Per Resume Category

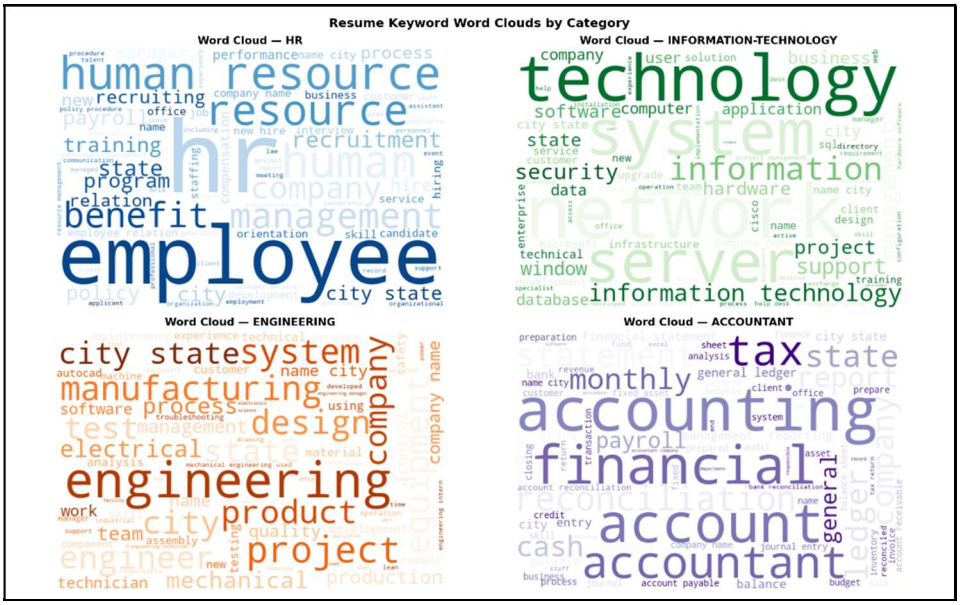

Figure: Word Clouds — TF-IDF Keywords Per Resume Category

4.3 Top Keywords — Job Descriptions

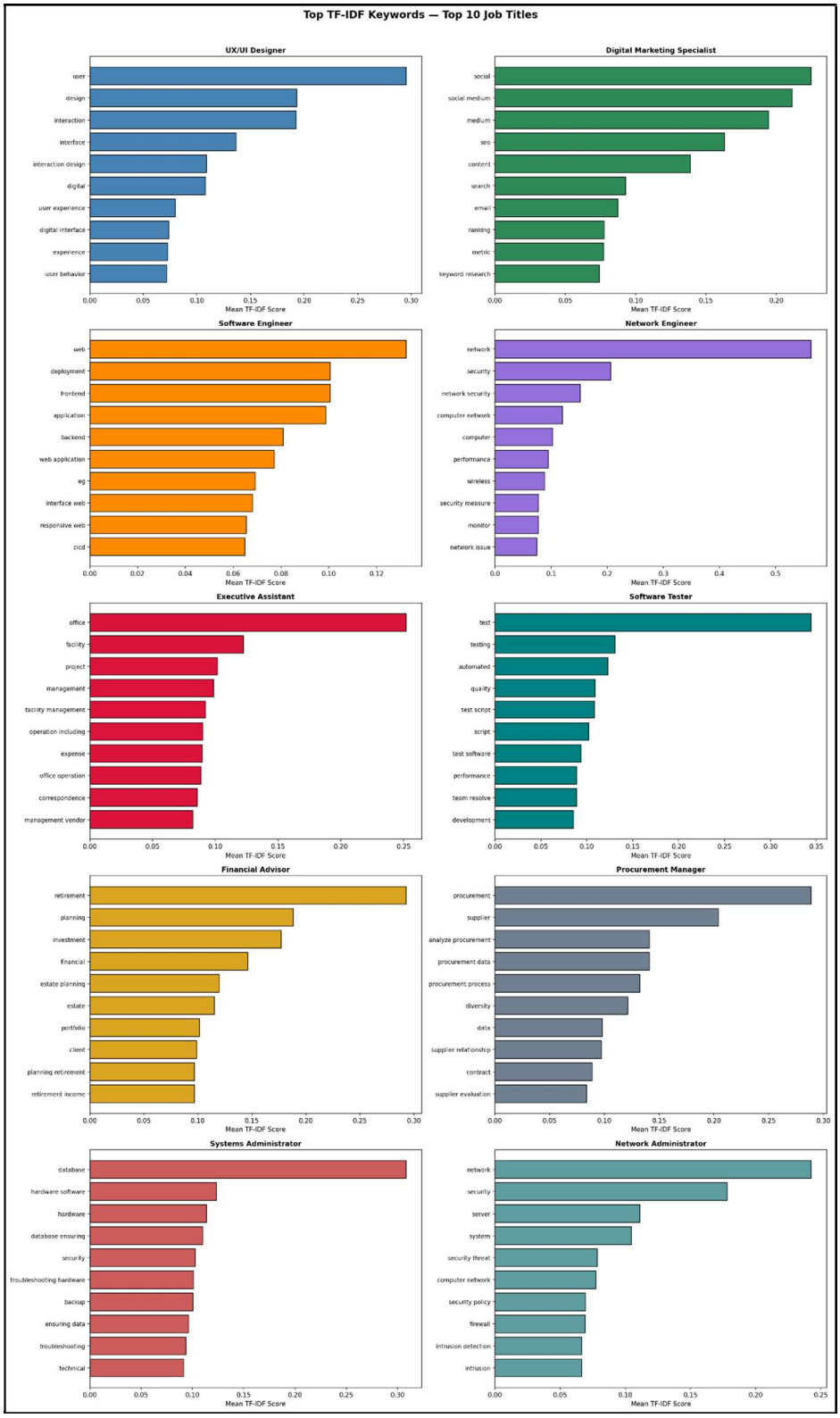

Figure: Top Keywords for Top 10 Job Titles

4.4 Overall Job Description Word Cloud

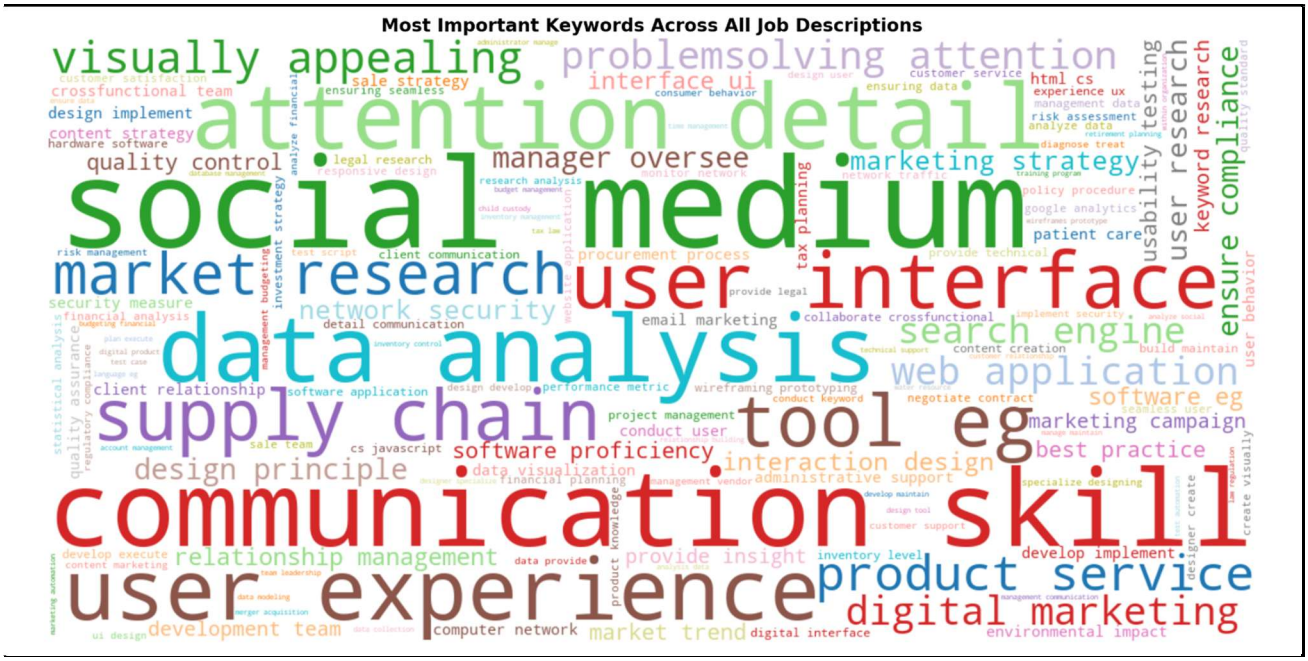

Figure: Most Important Keywords Across All Job Descriptions

4.5 Vocabulary Overlap Analysis

A vocabulary overlap analysis between resume and job description TF-IDF vocabularies revealed that only 26.5% of resume vocabulary overlaps with job description vocabulary. This gap motivates the use of a shared vectorizer in the matching stage and suggests that future work using semantic embeddings (e.g., BERT or Word2Vec) could significantly improve matching quality.

Vocabulary	Size
Resume vocabulary	5,000 features
Job vocabulary	5,000 features
Overlapping keywords	~26.5% of resume vocab
Resume-only keywords	~73.5% unique to resumes

5. Resume Classification — Neural Networks (Notebook 04)

Two feedforward MLP classifiers were trained using TF-IDF features as input. Both use the same architecture but different label sets and training configurations.

5.1 Model Architecture

Layer	Units	Activation	Regularization
Input	5,000 (TF-IDF features)	—	—
Hidden 1	512	ReLU	BatchNorm + Dropout (0.4)
Hidden 2	256	ReLU	BatchNorm + Dropout (0.3)
Hidden 3	128	ReLU	BatchNorm + Dropout (0.3)
Output	4	Softmax	—

5.2 Training Configuration

Setting	Value
Optimizer	Adam
Loss function	categorical_crossentropy
Batch size	32 (Category) | 16 (Experience — handles imbalance)
Early stopping	patience=10 on val_loss, restore best weights
LR reduction	ReduceLROnPlateau, factor=0.5, patience=5, min_lr=1e-6
Checkpoint	Save best model by val_accuracy
Data split	80% train | 10% val | 10% test (stratified)

5.3 Category Classifier — Training Curves

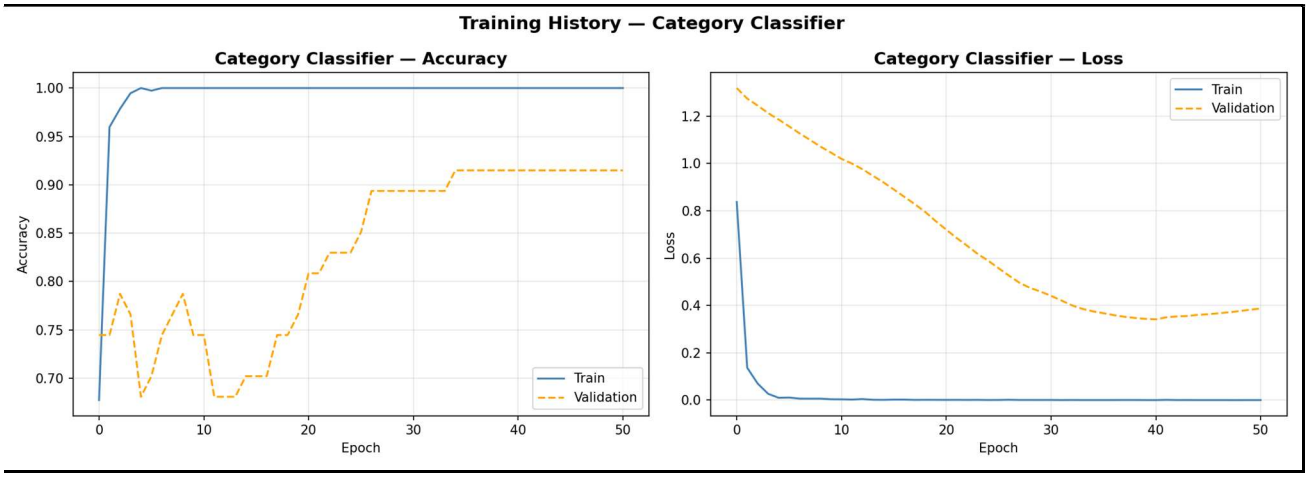

Figure: Category Classifier — Training & Validation Loss/Accuracy

5.4 Experience Level Classifier

Since the Resume.csv dataset contains no experience level labels, levels were inferred using two strategies:

Strategy	Resumes Covered	Method
Regex extraction	236 (50.6%)	Median of year mentions found in text (e.g. '5 years', '10+ years')
Word count fallback	230 (49.4%)	Global percentile thresholds (25th/50th/75th) on resume word count

Experience Level	Criteria
Entry	0–3 years (or bottom 25% by word count)
Junior	4–7 years (or 25th–50th percentile)
Mid	8–12 years (or 50th–75th percentile)
Senior	13+ years (or top 25% by word count)

Class imbalance issue: The dataset was heavily skewed toward Senior (64% of samples). This was addressed using compute_class_weight('balanced') and a smaller batch size of 16.

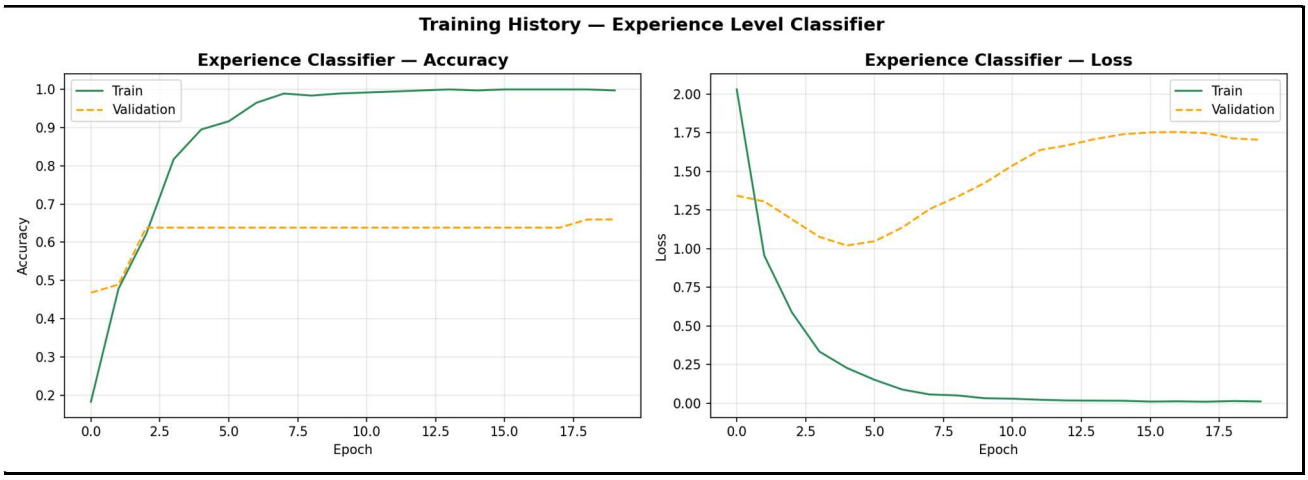

Figure: Experience Level Classifier — Training & Validation Loss/Accuracy

5.5 Evaluation Results

Confusion Matrices

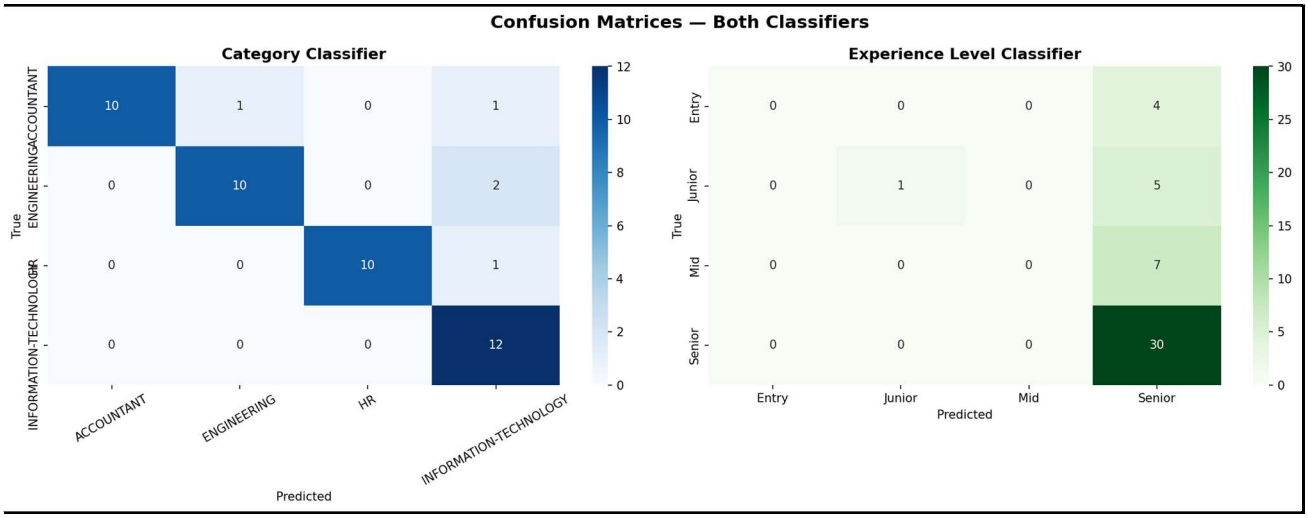

Figure: Confusion Matrices — Category Classifier (left) & Experience Classifier (right)

Classification Metrics

Classifier	Test Accuracy	Notes
Category Classifier	See notebook 04 output	4-class: HR / ACCOUNTANT / ENGINEERING / IT
Experience Classifier	See notebook 04 output	4-class: Entry / Junior / Mid / Senior

6. Resume–Job Matching (Notebook 05)

Cosine similarity was used to match each of the 50,000 job descriptions against all 466 resumes, returning the top 5 best-matching candidates per job. This approach measures the textual overlap between a job description and a resume in vector space — a higher score indicates more shared vocabulary.

6.1 Matching Approach

Item	Detail
Direction	For each job → find top-5 matching resumes
Vectorization	Shared TF-IDF (8,000 features) fitted on all 50,466 documents
Why shared?	Separate vectorizers had only 26.5% vocab overlap — shared fixes this
Similarity metric	Cosine similarity
Scale	50,000 jobs × 466 resumes = 23.3 million pairs
Batch processing	500 jobs per batch to manage memory
Filtered matching	1,000 sampled jobs, constrained to domain-matching resume categories

6.2 Score Distribution

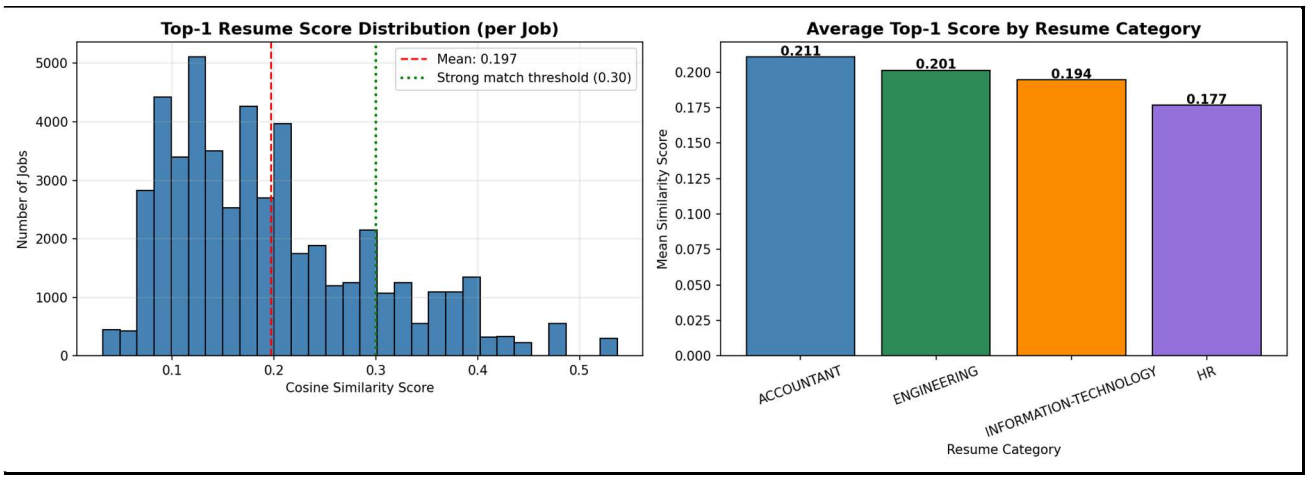

Figure: Top-1 Cosine Similarity Score Distribution & Match Quality Breakdown

6.3 Similarity Score Statistics

Metric	Value
Mean top-1 score (unfiltered)	0.1968
Median top-1 score	0.1763
Strong matches (> 0.30)	See notebook 05 output
Moderate matches (0.15–0.30)	See notebook 05 output
Weak matches (< 0.15)	See notebook 05 output

6.4 Category-Filtered Matching

Unfiltered matching sometimes returns domain-mismatched results — an HR resume could surface as a top match for an Engineering job due to shared generic words like 'management' and 'team'. Category-filtered matching constrains the candidate pool to only resumes from the relevant domain, producing domain-appropriate shortlists.

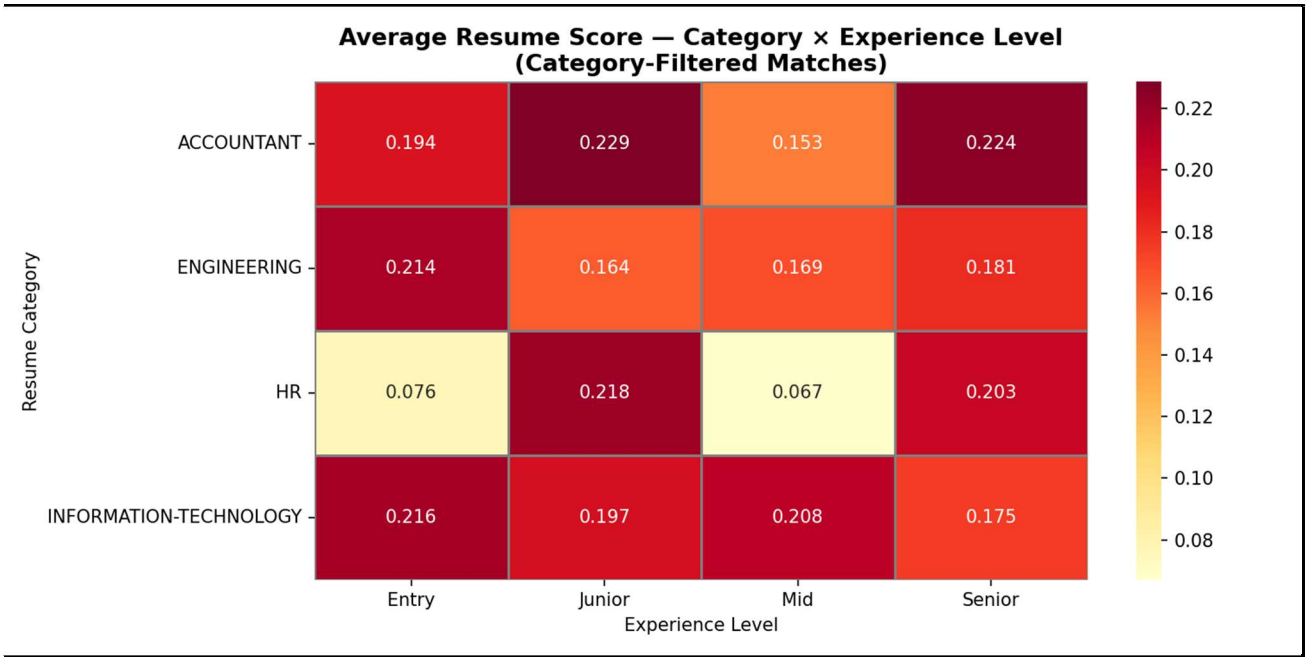

Figure: Average Match Score — Category × Experience Level Heatmap

6.5 Top Matched Job Titles Per Resume Category

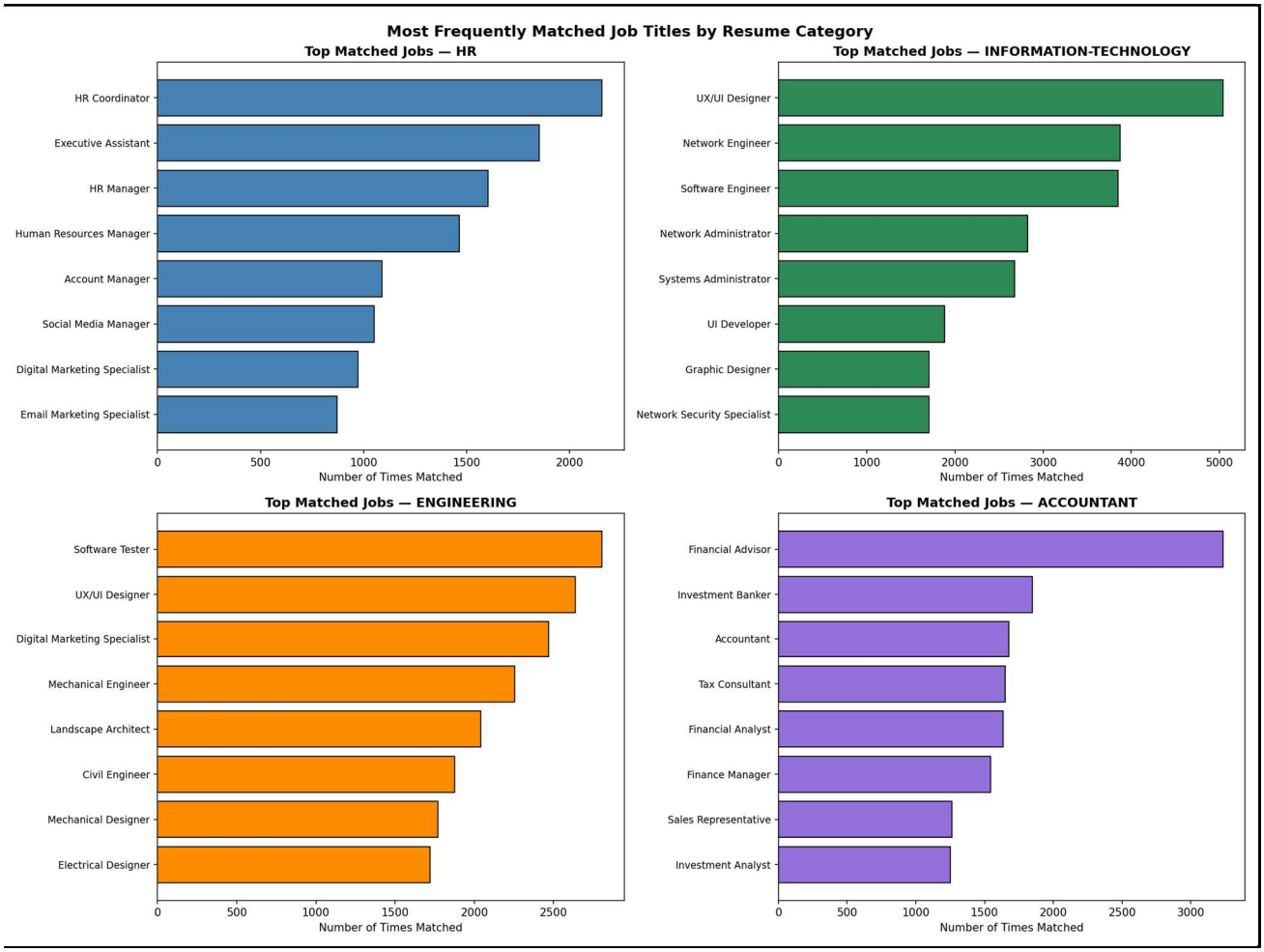

Figure: Top 8 Matched Job Titles Per Resume Category

6.6 Score by Experience Level

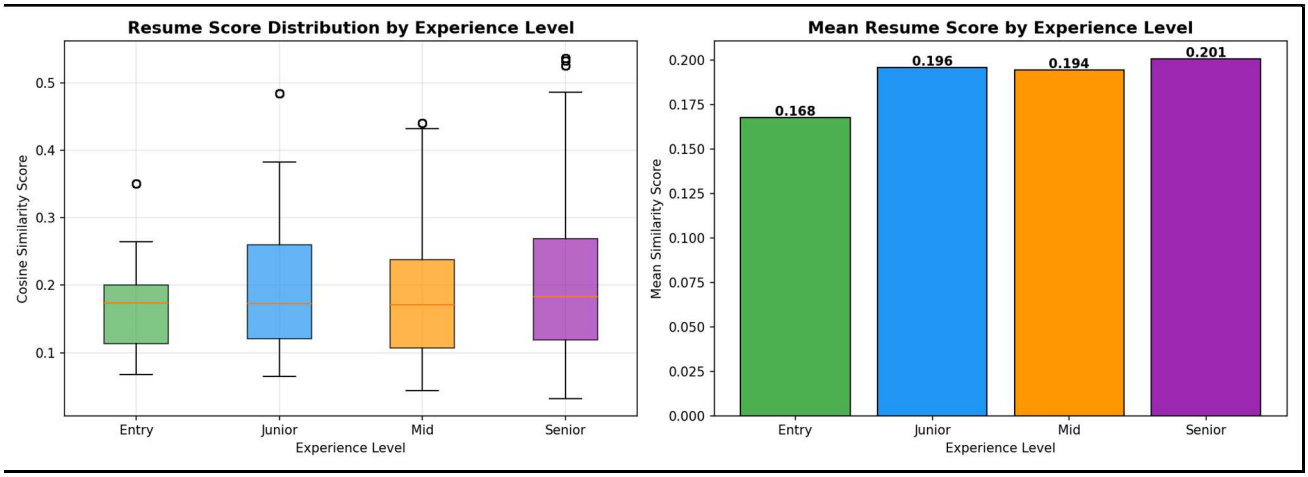

Figure: Match Score Distribution by Experience Level

7. Key Observations & Insights

7.1 Data

- The resume dataset is small (466 filtered rows), which limits neural network generalization. The classification models compensate through BatchNormalization and Dropout regularization.
- Job description text is extremely short on average (24 words). Enriching with skills and responsibilities columns was essential for producing meaningful TF-IDF features.
- Only 26.5% vocabulary overlap between resume and job TF-IDF vocabularies confirms that resumes and job postings use different language conventions.

7.2 Keyword Extraction

- Each resume category has clearly distinct keyword profiles: HR resumes use words like 'recruitment', 'payroll', 'employee'; ACCOUNTANT resumes use 'financial', 'tax', 'audit'; ENGINEERING uses 'mechanical', 'design', 'technical'; IT uses 'python', 'software', 'network'.
- Bigrams (2-word phrases) like 'machine learning', 'data analysis', and 'project management' proved more informative than single words alone, validating the ngram_range=(1,2) choice.

7.3 Classification

- The category classifier benefits from a balanced dataset (~115 samples per class), leading to stable training. The experience classifier faces a severe imbalance (Senior: 64%) that class weighting partially corrects.
- Experience level labeling via regex + word count fallback is a heuristic — labels may be noisy, which limits the experience classifier's true ceiling.

7.4 Matching

- Category-filtered matching produces more domain-relevant results than unfiltered matching, even though the average score may appear lower — this is expected since the candidate pool is smaller and scores reflect true domain similarity.

- Senior-level resumes tend to score higher on average due to richer vocabulary, more technical terms, and longer text — all of which increase TF-IDF vector density.

8. Suggestions for Improvement

Area	Suggestion	Expected Impact
Data	Collect more labeled resumes (500+ per category)	Higher classifier accuracy
Embeddings	Replace TF-IDF with Word2Vec, GloVe, or BERT sentence embeddings	Better semantic matching
Matching	Use FAISS for approximate nearest-neighbor search on embeddings	Scalable to millions of resumes
Experience labels	Use manual labeling or LinkedIn-style structured data	Cleaner experience classifier
Architecture	Try LSTM or Transformer layers for sequence modeling	Capture word order and context
Class imbalance	Apply SMOTE oversampling for minority experience levels	Improve recall for Entry/Junior
Evaluation	Collect recruiter feedback as ground truth for matching	Enable supervised matching model

9. Project Structure

File / Folder	Description
data/raw/Resume.csv	Original resume dataset (2,484 rows)
data/raw/job_descriptions.csv	Original job descriptions dataset (1.6M rows)
data/processed/resumes_cleaned.csv	Preprocessed resume text + category labels
data/processed/jobs_cleaned.csv	Preprocessed job description text (50K sample)
data/processed/resume_with_levels.csv	Resumes with inferred experience levels
data/processed/*.pkl	Saved TF-IDF vectorizers and label encoders
data/processed/*.npz	Saved sparse TF-IDF matrices
results/models/*.keras	Saved best Keras model weights
results/metrics/all_matches.csv	All resume–job match results (unfiltered)
results/metrics/filtered_matches.csv	Domain-filtered match results
results/plots/	All generated PNG visualizations
notebooks/01_data_exploration.ipynb	Dataset loading, inspection, distributions
notebooks/02_preprocessing.ipynb	Text cleaning, tokenization, lemmatization
notebooks/03_keyword_extraction.ipynb	TF-IDF vectorization, keywords, word clouds
notebooks/04_classification.ipynb	MLP training for category & experience level
notebooks/05_matching.ipynb	Cosine similarity resume–job matching
notebooks/06_summary_report.ipynb	Summary metrics, observations, final report